

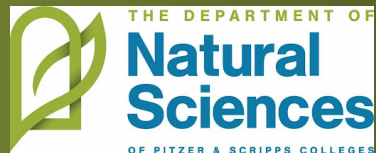
Applying Maximum Entropy Models to Organism Sensors

MAA MathFest 2026, Boston, MA

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Motivation

How do biological organisms understand their environment?



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How do biological organisms understand their environment?

1. Encodes information about environment
2. Past, present, and future
3. Constraint: energetic cost of encoding



Entropy: a Fundamental Player

How well can we encode/compress information?

Shannon Entropy

For a random variable X and event space $\mathcal{X}$,

$$H[X] = - \sum_{x \in \mathcal{X}} p(x) \log_b(p(x)), \quad (1)$$

where $b = 2, e, 10$ most commonly.

Entropy: a Fundamental Player

Flipping Coins ($b = 2$)

Entropy of one fair coin flip (C) and one biased (B , $p_{\text{heads}} = 0.7$):

$$H[C] = - \sum_{x \in \mathcal{X}} p(x) \log_b(p(x)) = -\log_2(1/2) = 1 \text{ bit.} \quad (2)$$

$$H[B] = -0.7 \log_2(0.7) - 0.3 \log_2(0.3) \approx 0.88 \text{ bits.} \quad (3)$$

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$$H[C_1, C_2] = -\log_2(1/4) = H[C] + H[C] = 2 \text{ bits.} \quad (4)$$

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$$H[C_1 + C_2] = 1.5 \text{ bits.} \quad (5)$$

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Key takeaways:

- $H[X, Y] \leq H[X] + H[Y]$, equality if X, Y ind.
- “Surprisal”: $-\log p$
- **Maximum entropy maximizes surprisal**

Mutual Information: One Take on Entropy

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where S is the environment/stimulus and X is the biological sensor system.

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- Typically used in the field
- Yields one loosely interpretable value given S, X
- Often difficult to calculate (many estimators are needed)
- Gives just one piece of information about sensor-environment system

Maximum Entropy: An Alternate Lens

THE GENERAL PIPELINE

- Given constraints, get $p(\mathbf{x})$ that maximizes $H[X]$ (maximal 'surprisal').
 - Least biased model
 - Equivalent to minimizing KL divergence
 - Stat mech models work too!

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$$p(\mathbf{x}) = \frac{1}{Z} \exp \left(\sum_{\mu} \lambda_{\mu} f_{\mu}(\mathbf{x}) \right) \quad (7)$$

- $\mathbf{x}$: Joint environment-sensor state vector
- Z : Normalization constant
- λ_{μ} : Constraint Lagrange multipliers
- f_{μ} : Constraint equations

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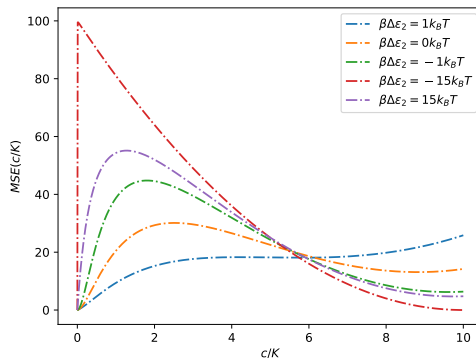
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 - $\hat{x}(s) = \arg \max_x p(x | s)$
- How good is the estimator?
 - Ordered: bias, variance
 - Categorical: confusion matrices

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MaxEnt: New Insights with Cooperative Sensors

- Two-sensor system: Cooperativity or not?
- Mutual information
 - No cooperativity is best
- MaxEnt
 - Depends on the ligand concentration
 - More aligned with previous findings



MaxEnt: More Information with Cultured Cortical Neurons

- Model probability distribution ($\mathbf{x} = (x, \mathbf{s})$):

$$p(\mathbf{x}) = \frac{1}{Z} \exp(\mathbf{x}^\top J \mathbf{x}) \quad (8)$$

- Estimator: compute likelihood ratio via

$$L(\mathbf{s}) = \exp(-J_{xx} - 2J_{xs} \cdot \mathbf{s}) \quad (9)$$

Predicted 0
Predicted 1

Actual 0

Actual 1

		Actual 0	Actual 1
Predicted 0		True negative rate	False negative rate
		False positive rate	True positive rate

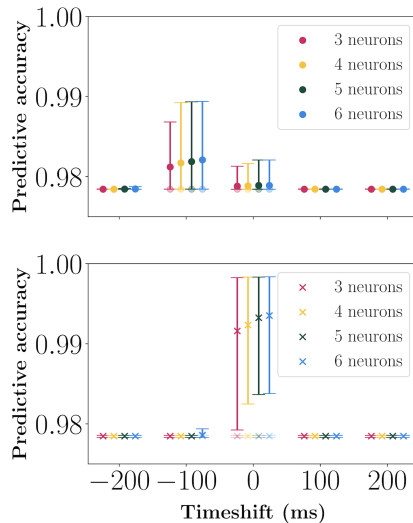
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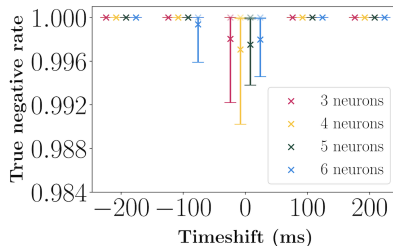
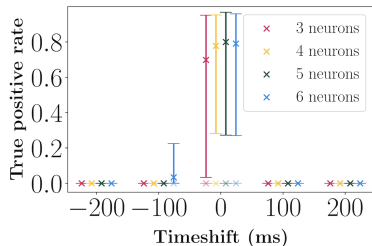
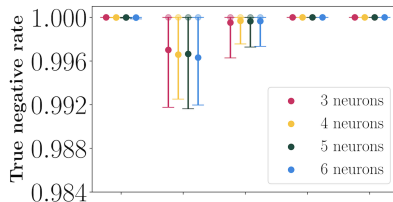
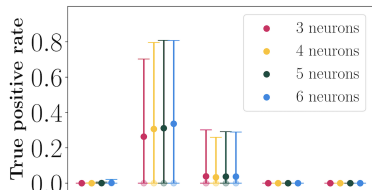
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MaxEnt: More Information with Cultured Cortical Neurons



Recap

	Mutual Information	Bias/Variance	Confusion Matrices
Variable Type	Any	Ordered / continuous	Categorical
Output	Single scalar	Bias, Variance as a function of input	Error rates per category
Interpretation	One loosely interpretable value	Accuracy (bias), precision (variance)	Per-class error breakdown
Computational Scaling	Exponential	Roughly power law	Power law or exponential, depending on situation

Acknowledgements

- PI: Sarah Marzen
- Co-authors: Eliana Schimke, Martina Lamberti, Joost le Feber, Tristan Kako, Sarah Marzen

Thank you!

Here's the paper for more info: <https://www.biorxiv.org/content/10.64898/2026.01.12.698979v1>

